PROBLEM 31

Two Particles on Cone with String

Original statement · Problems.pdf, p. 19

Multi-Particle Cable Constraint

Two heavy point particles P_1 and P_2 of mass m each are connected by an inextensible, massless string of total length ℓ. Particle P_1 is constrained to move without friction over the surface of a cone of semiangle α (the particle cannot leave the surface of the cone). The string passes through an aperture at the cone vertex O, and it can slide through it without friction. Particle P_2 hangs along the cone axis, and is constrained to move along it. We define an inertial reference frame S_0 as shown in the figure. We call ξ the length of the fraction of string between O and P_1, and θ the angle formed between the Ox axis and the projection onto the Oxy plane of the $\mathbf{OP}_1$ vector. In the following, assume that the string is always tense and that the particles do not reach point O during motion.

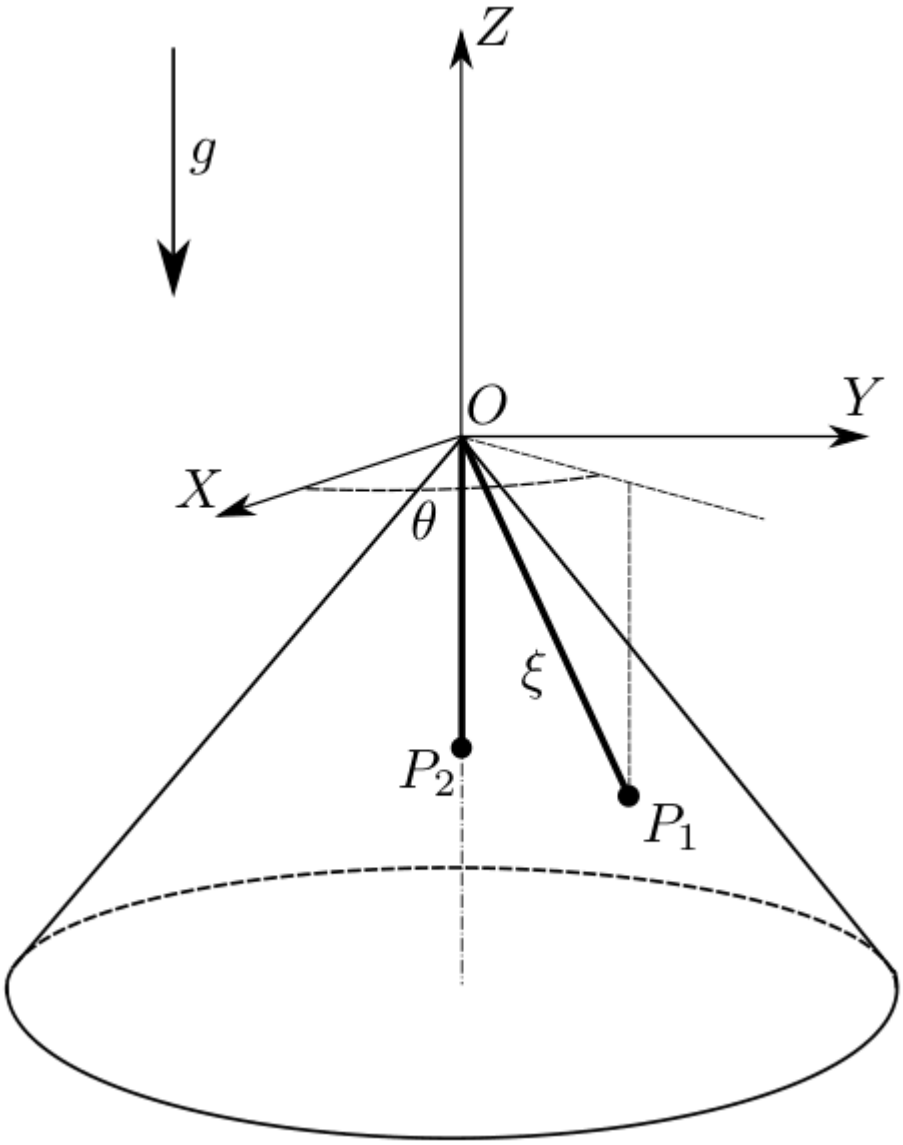


Figure: Cone with vertex O and axes X, Y, Z; P1 on the cone surface at string length ξ and angle θ, P2 hanging along the axis (Problems.pdf, p. 19).

(a) Write down the general expression of the position, velocity and acceleration vectors of P_1 and P_2 as a function of θ, ξ and the parameters of the problem.

(b) Write down the expression of all forces that act on P_1 and P_2.

(c) Write down the differential equations that, if solved, would yield the time evolution of θ, ξ, the tension on the string, and any other reaction force in the problem.

(d) What is the mathematical condition that the function $\xi(t)$ must satisfy for the string to remain tense all the time?

(e) How fast should P_1 rotate in the θ direction to have a circular trajectory over a cone parallel? What is the value of the string tension and any other reaction force in the problem in that case? Give your answer as a function of ξ and the parameters of the problem.

(f) Taking these equations as the starting point (or based on the physical considerations of the problem) write down two first-integrals of the problem that do not depend on the string tension or any other reaction force in the problem.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** two heavy particles P_1 and P_2 of mass m are joined by an inextensible massless string of length ℓ that passes without friction through the aperture at the vertex O of a cone of semi-angle α. P_1 moves without friction on the cone surface; P_2 hangs along the cone axis (Oz vertical upward, the cone opens downward). The string stays taut and the particles do not reach O.
- **Constraints:** with 6 coordinates for two free particles:
  - P_1 on the cone: 1 constraint (bilateral: the statement says the particle cannot leave the surface).
  - Constant string length: $\xi + (\text{depth of } P_2) = \ell$, i.e. 1 constraint (unilateral: $T \geq 0$).
  - P_2 on the axis: $x_2 = y_2 = 0$, 2 constraints.
- **Degrees of freedom:** $\text{CDOF} = 6 - 4 = 2$, with ξ (the length of the string between O and P_1) and θ (the azimuthal angle of P_1).
- **Parameters:** m, ℓ, α, g.

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{k}$ vertical upward.
- **Cylindrical basis:** $\mathbf{e}_R = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$, $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$, $\mathbf{k}$ (rotation about $\mathbf{k}$ by θ, matrix $\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det = \cos^2\theta + \sin^2\theta = 1$; RR^T has diagonal entries $\cos^2\theta + \sin^2\theta = 1$ (and 1) and off-diagonal entries $\cos\theta\sin\theta - \sin\theta\cos\theta = 0$, so $RR^T = I$; $\dot{\mathbf{e}}_R = \dot\theta\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_R$).
- **Cone basis:** the unit vector along the generator from O to P_1, $\mathbf{e}_\xi = \sin\alpha\,\mathbf{e}_R - \cos\alpha\,\mathbf{k}$; the unit outward normal to the cone, $\mathbf{n} = \cos\alpha\,\mathbf{e}_R + \sin\alpha\,\mathbf{k}$; and $\mathbf{e}_\theta$. The triple $\{\mathbf{e}_\xi, \mathbf{e}_\theta, \mathbf{n}\}$ is right-handed ($\mathbf{e}_\xi \times \mathbf{e}_\theta = \mathbf{n}$). In the cylindrical basis $\{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{k}\}$

$$[{}_1R_2] = \begin{pmatrix} \sin\alpha & 0 & \cos\alpha \\ 0 & 1 & 0 \\ -\cos\alpha & 0 & \sin\alpha \end{pmatrix}, \qquad \det[{}_1R_2] = \sin\alpha\cdot\sin\alpha + \cos\alpha\cdot\cos\alpha = \sin^2\alpha + \cos^2\alpha = 1$$

and its columns have unit norm ($\sin^2\alpha + \cos^2\alpha = 1$) and are orthogonal ($\sin\alpha\cos\alpha - \cos\alpha\sin\alpha = 0$), so $[{}_1R_2][{}_1R_2]^T = I$.

PHASE 3

Derivation

3.1 Kinematics

(a) Position, velocity and acceleration

Pedagogical Justification: P_1 is at the distance ξ from O along the generator, so $\mathbf{r}_1 = \xi\,\mathbf{e}_\xi = \xi\sin\alpha\,\mathbf{e}_R - \xi\cos\alpha\,\mathbf{k}$. Since $\mathbf{e}_R$ rotates with θ, the derivatives use the product rule and Poisson's formula. P_2 is on the axis at the depth $\ell - \xi$ below O (the string has length ξ on the cone side and $\ell - \xi$ on the hanging side).

$$\mathbf{v}_1 = \dot\xi\,\mathbf{e}_\xi + \xi\sin\alpha\,\dot\theta\,\mathbf{e}_\theta$$

From $\mathbf{r}_1 = \xi\,\mathbf{e}_\xi$ with $\dot{\mathbf{e}}_\xi = \sin\alpha\,\dot\theta\,\mathbf{e}_\theta$: $\mathbf{v}_1 = \dot\xi\,\mathbf{e}_\xi + \xi\dot\theta\sin\alpha\,\mathbf{e}_\theta$. Differentiating again (product rule) with $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_R$ and $\mathbf{e}_R = \sin\alpha\,\mathbf{e}_\xi + \cos\alpha\,\mathbf{n}$: $\mathbf{a}_1 = \ddot\xi\,\mathbf{e}_\xi + 2\dot\xi\dot\theta\sin\alpha\,\mathbf{e}_\theta + \xi\ddot\theta\sin\alpha\,\mathbf{e}_\theta - \xi\dot\theta^2\sin\alpha\,(\sin\alpha\,\mathbf{e}_\xi + \cos\alpha\,\mathbf{n})$, i.e.

$$\mathbf{a}_1 = \left(\ddot\xi - \xi\dot\theta^2\sin^2\alpha\right)\mathbf{e}_\xi + \sin\alpha\left(\xi\ddot\theta + 2\dot\xi\dot\theta\right)\mathbf{e}_\theta - \xi\dot\theta^2\sin\alpha\cos\alpha\,\mathbf{n}$$

$$\mathbf{r}_2 = -(\ell - \xi)\,\mathbf{k}, \qquad \mathbf{v}_2 = \dot\xi\,\mathbf{k}, \qquad \mathbf{a}_2 = \ddot\xi\,\mathbf{k}$$

3.2 Forces

(b) Forces on P_1 and P_2

- **On P_1:** the weight $-mg\,\mathbf{k} = mg\cos\alpha\,\mathbf{e}_\xi - mg\sin\alpha\,\mathbf{n}$; the tension of the string, directed toward O, $\mathbf{T}_1 = -T\,\mathbf{e}_\xi$ ($T > 0$); and the reaction of the cone $\mathbf{N} = N\,\mathbf{n}$ (outward).
- **On P_2:** the weight $-mg\,\mathbf{k}$; the tension $+T\,\mathbf{k}$ (same string, massless, smooth aperture); and the reaction $\mathbf{R}_2$ of the guide, perpendicular to $\mathbf{k}$ (horizontal).
- **Inertia forces:** none (S_0 is inertial).

3.3 Equations of Motion

(c) Differential equations

> **Pedagogical Justification:** Newton's second law for each particle, six scalar equations for the six unknowns ξ, θ, T, N and the two horizontal components of $\mathbf{R}_2$:

$$P_2,\ \mathbf{k}:\ m\ddot\xi = T - mg \implies T = m\left(g + \ddot\xi\right), \qquad P_2,\ \text{horizontal}:\ \mathbf{R}_2 = \mathbf{0}\ (\text{since } \mathbf{a}_2 \parallel \mathbf{k})$$

$$P_1,\ \mathbf{e}_\xi:\ m\left(\ddot\xi - \xi\dot\theta^2\sin^2\alpha\right) = mg\cos\alpha - T$$

$$P_1,\ \mathbf{e}_\theta:\ m\sin\alpha\left(\xi\ddot{\theta}+2\dot{\xi}\dot{\theta}\right)=0, \qquad P_1,\ \mathbf{n}:\ -m\xi\dot{\theta}^2\sin\alpha\cos\alpha=N-mg\sin\alpha$$

- **Decoupling.** Substituting $T=m(g+\ddot{\xi})$ in the $\mathbf{e}_\xi$ equation eliminates the tension and leaves the equations of the two free coordinates:

$$\boxed{2\ddot{\xi}-\xi\dot{\theta}^2\sin^2\alpha+g\left(1-\cos\alpha\right)=0}, \qquad \boxed{\xi\ddot{\theta}+2\dot{\xi}\dot{\theta}=0}$$

and, once $\xi(t)$ and $\theta(t)$ are known,

$$T=m\left(g+\ddot{\xi}\right), \qquad N=mg\sin\alpha-m\xi\dot{\theta}^2\sin\alpha\cos\alpha$$

- **First integrals** (independent of T and N): see 3.4(f).

3.4 Integration and Answers

(d) Condition for the string to remain taut

Pedagogical Justification: the string can only pull, so $T>0$ at all times:

$$T=m\left(g+\ddot{\xi}\right)>0 \iff \boxed{\ddot{\xi}>-g}$$

Substituting $\ddot{\xi}=\frac{1}{2}\left[\xi\dot{\theta}^2\sin^2\alpha-g(1-\cos\alpha)\right]$ from the decoupled equation of (c) gives $T=\frac{m}{2}\left[g\left(1+\cos\alpha\right)+\xi\dot{\theta}^2\sin^2\alpha\right]>0$ for $0<\alpha<\pi$, so the string stays taut throughout the motion.

Discrepancy with the official solution
The key of `Problems.pdf` states the condition as $\ddot{\xi}>g$. That cannot be right: for the horizontal circular trajectory of part (e) the key itself gives $T=mg>0$ with $\ddot{\xi}=0$, which violates $\ddot{\xi}>g$, while $\ddot{\xi}=0>-g$ is satisfied. The equation of P_2 gives $T=m(g+\ddot{\xi})$, hence $\ddot{\xi}>-g$ (the minus sign is missing in the key).

(e) Circular trajectory on a parallel

> **Pedagogical Justification:** a horizontal circle on the cone has ξ constant, so $\dot{\xi} = \ddot{\xi} = 0$ and the radial equation reduces to

$$-\xi\dot{\theta}^2 \sin^2\alpha + g\,(1-\cos\alpha) = 0 \implies \boxed{\dot{\theta} = \frac{1}{\sin\alpha}\sqrt{\frac{g\,(1-\cos\alpha)}{\xi}}}$$

- **Tension:** $T = m\,(g+0) = mg$, directed on P_1 as $\mathbf{T}_1 = -mg\,\mathbf{e}_\xi = mg\,(-\sin\alpha\,\mathbf{e}_R + \cos\alpha\,\mathbf{k})$.
- **Cone reaction:** with $\xi\dot{\theta}^2\sin^2\alpha = g\,(1-\cos\alpha)$:

$$N = mg\sin\alpha - m\cos\alpha\,\frac{g\,(1-\cos\alpha)}{\sin\alpha} = \frac{mg}{\sin\alpha}\left(\sin^2\alpha - \cos\alpha + \cos^2\alpha\right) = \boxed{\frac{mg\,(1-\cos\alpha)}{\sin\alpha}}, \qquad \mathbf{N} = N\,(\cos\alpha\,\mathbf{e}_R + \sin\alpha\,\mathbf{k})$$

$N > 0$ for $0 < \alpha < \pi$: the cone presses on the particle (which would also be admissible for a one-sided contact).

(f) Two first integrals independent of the tension and the reactions

- **Azimuthal integral (first integral of the $\mathbf{e}_\theta$ equation).** With the product rule, the azimuthal acceleration is $\sin\alpha\left(\xi\ddot{\theta} + 2\dot{\xi}\dot{\theta}\right) = \dfrac{\sin\alpha}{\xi}\dfrac{d}{dt}\left(\xi^2\dot{\theta}\right)$. Gravity, the tension (toward O) and the cone reaction all lie in the meridian plane of P_1, so they have no $\mathbf{e}_\theta$ component:

$$\frac{d}{dt}\left(\xi^2\dot{\theta}\right) = 0 \implies \boxed{\xi^2\dot{\theta} = C_1}$$

- **Energy.** The reactions do no work: $\mathbf{N}\cdot\mathbf{v}_1 = 0$ ($\mathbf{N}$ is normal to the surface) and the tension does the work $-T\dot{\xi}$ on P_1 and $+T\dot{\xi}$ on P_2, which cancel (inextensible string, smooth aperture). Gravity is conservative. The kinetic and potential energies are

$$K = \tfrac{1}{2}m\left(\dot{\xi}^2 + \xi^2\dot{\theta}^2\sin^2\alpha\right) + \tfrac{1}{2}m\dot{\xi}^2 = m\dot{\xi}^2 + \tfrac{1}{2}m\xi^2\dot{\theta}^2\sin^2\alpha, \qquad V = -mg\xi\cos\alpha - mg(\ell-\xi) = mg\xi\,(1-\cos\alpha) - mg\ell$$

$$\boxed{2\dot{\xi}^2 + \xi^2\dot{\theta}^2\sin^2\alpha + 2g\xi\,(1-\cos\alpha) = C_2}$$

where the constant C_2 absorbs $-2g\ell$ and the factor $2/m$.

PHASE 4

Interpretation & Checks

- **Effective potential.** With $\xi^2\dot{\theta} = C_1$ the energy becomes $2\dot{\xi}^2 + W(\xi) = C_2$ with $W(\xi) = \dfrac{C_1^2\sin^2\alpha}{\xi^2} + 2g\xi\,(1-\cos\alpha)$. Its minimum is at $W'(\xi_c) = -\dfrac{2C_1^2\sin^2\alpha}{\xi_c^3} + 2g\,(1-\cos\alpha) = 0$, i.e. $\xi_c\dot{\theta}^2\sin^2\alpha = g\,(1-\cos\alpha)$, the circular parallel of (e); and $W'' = \dfrac{6C_1^2\sin^2\alpha}{\xi^4} > 0$, so it is a stable minimum.
- **Dimensional check:** $[\xi^2\dot{\theta}] = \mathrm{m^2 s^{-1}}$ and $[g\xi] = \mathrm{m^2 s^{-2}} = [\dot{\xi}^2]$; $[T] = [N] = \mathrm{N}$.
- **Check against the official key:** the key of `Problems.pdf` gives two degrees of freedom, the quadratures $\xi^2\dot{\theta} = C_1$ and $2\dot{\xi}^2 + \xi^2\dot{\theta}^2\sin^2\alpha + 2g\xi(1-\cos\alpha) = C_2$, the circular condition $\xi\dot{\theta}^2\sin^2\alpha = g(1-\cos\alpha)$, $\mathbf{T} = mg\,(-\sin\alpha\,\mathbf{e}_R + \cos\alpha\,\mathbf{k})$ and $\mathbf{N} = \dfrac{mg(1-\cos\alpha)}{\sin\alpha}(\cdots)$. Everything agrees with the results above, with two remarks: the tension condition of the key ($\ddot{\xi} > g$) is an erratum (see (d)), and the key writes the direction of $\mathbf{N}$ with $\mathbf{e}_\theta$, which is tangent to the parallel and cannot be normal to the cone; the normal direction is $\cos\alpha\,\mathbf{e}_R + \sin\alpha\,\mathbf{k}$. Sympy confirms the three components of $\mathbf{a}_1$ and numerical integration of the equations of (c) ($\alpha = 40^\circ$) conserves $\xi^2\dot{\theta}$ and the energy to 3×10^{-14}, with T and N positive throughout and $N/(mg) = 0.363970$ on the circular orbit, equal to $(1-\cos\alpha)/\sin\alpha$.

Source: Problems.pdf